The European Commission's science and knowledge service

Joint Research Centre

VECTO Trailer Tool



European Commission

Content

- References to information in Technical Annexes of the Trailer Regulation
- VECTO Trailer Tool
  - Software package
  - JobFiles GUI
  - Options GUI
  - Background worker
  - How to create or edit trailer job
  - Result files
- Generation of input XML for a certified aero device
- Overview content of VECTO Trailer Tool Masterexcel

European Commission

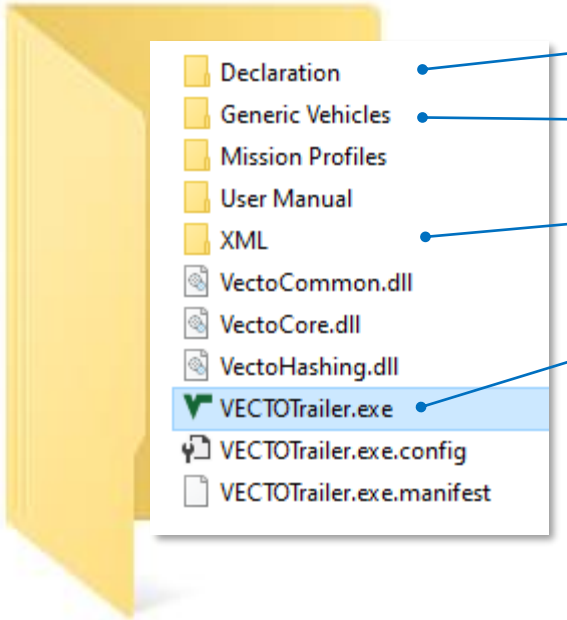
References to information in Technical Annexes of the Trailer Regulation

Annex Number	Title	Further Explanation
Annex I	Classification of vehicles in vehicle groups	Required definitions and classification system for trailers covered by the Regulation
Annex II	Requirements and processes for the operation of the simulation tool	Requirements for the process to be able to determine the official values within the framework of trailer regulation by means of the VECTO Trailer Tool
Annex III	Input information about the characteristic of the vehicle	Definitions and precise description of the input parameters into the tool and how to determine them
Annex IV	Template of the Manufacturer‘s Records file and of the Customer Information file	Definition of content and structure of the Manufacturer‘s Records file (MRF) and the Customer Information file (CIF)
Annex V	Vehicle‘s air drag data	Defines the procedures how the features of aerodynamic devices shall be determined, either via virtual testing using CFD or by applying standard reduction rates for aerodyanmic devices fulfilling certain minimum properties

European Commission

VECTO Trailer Tool – Software package

Content of VECTO Trailer Tool package as downloaded from JRC server can be copied to any place on a PC/server*



Hard coded data via *.csv files for information purposes

Example files for trailers and tyres and XML files of the generic towing vehicles and certified aerodynamic devices

Example XML files and XSD files

Executable, just double-click to start

* Snapshot shows only a reduced set of data included in the downloadable package

European Commission

VECTO Trailer Tool – JobFiles GUI

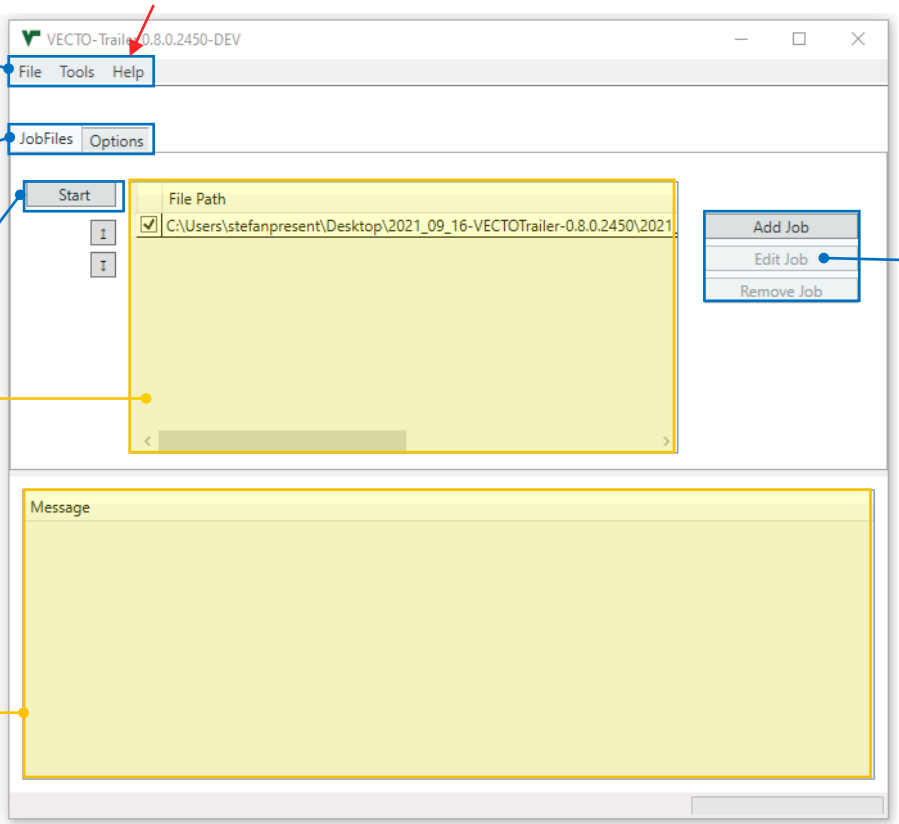
Toolbar with common controls like “add new Job”

Switch between “Job Files” and “Options” tab

Start button to simulate active Jobs

Job window that shows all loaded Job files

Message window showing the calculation status



Buttons to handle Job files

- Add new Jobs
- Edit existing Jobs from Job window
- Delete Jobs from Job window

European Commission

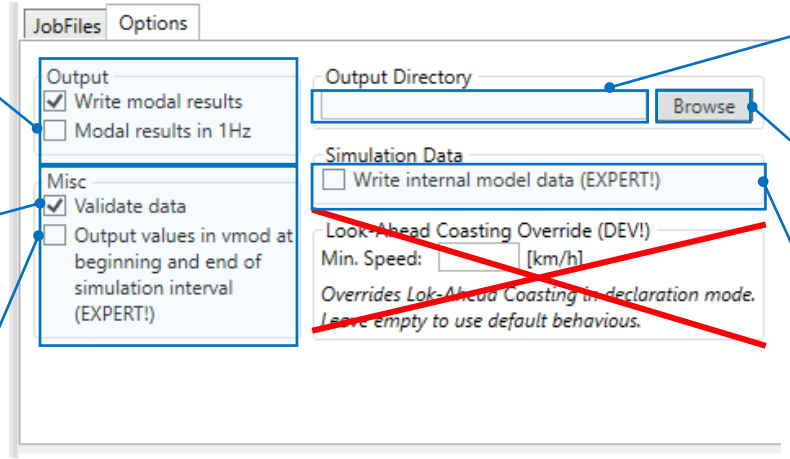
VECTO Trailer Tool – Options GUI

- At first program start it is recommended to check the main tool settings and define the output directory
- The settings displayed below are the default, recommended settings
- Settings only affect the handling of input data or results, but do not influence the official results.

Modal results in 1 Hz:
If selected, the modal results (.vmod file) will be converted into 1Hz after the simulation. This may add certain artefacts in the resulting modal results file.

Enables or disables internal checks if the model parameters are within a reasonable range.

If this option is enabled, the .vmod file will contain two entries for every simulation interval, one at the beginning and one at the end of the simulation interval.



Output directory

Browse button

Checkbox to create JSON file with input/simulation data of simulated Jobs

European Commission

VECTO Trailer Tool - Background worker

Enables a simple autonomisation of simulation runs

1. The background worker periodically searches a directory specified by the user for new input XMLs
2. If a new file is available, a calculation is automatically started in the background
3. After completion, the result files are stored in another directory to be specified by the user

European Commission

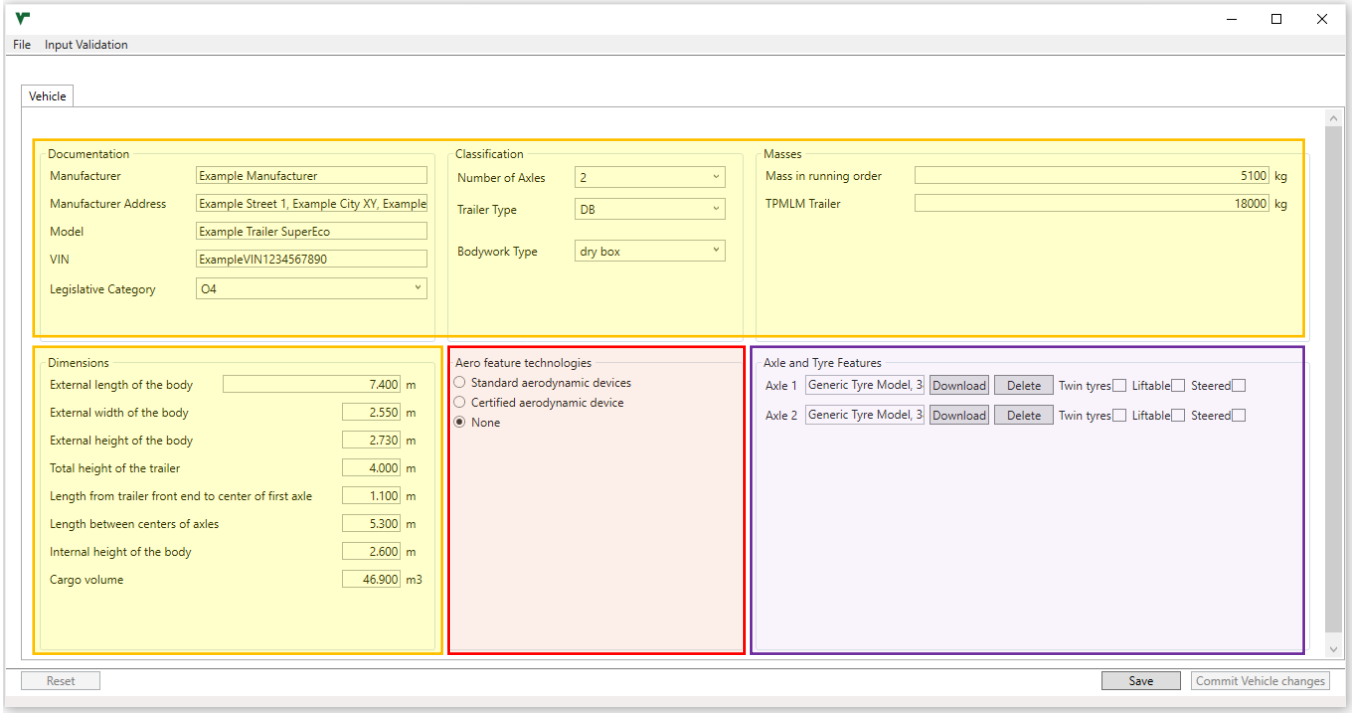
VECTO Trailer Tool – Create or edit trailer job (1/6)

1. Open window via
   - “Edit Job” or
   - Right click on Job window → “Create Trailer Job”

2. Define trailer specifications

3. Define Aero feature technologies

4. Define Axle and Tyre features



European Commission

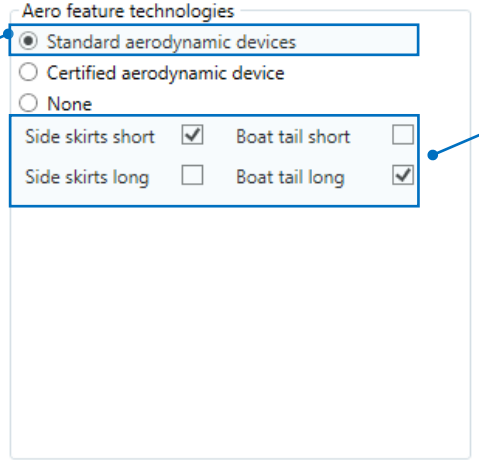
VECTO Trailer Tool – Create or edit trailer job (2/6)

3. Define Aero feature technologies

Choose this if the trailer is equipped with standard aerodynamic device(s)

Reduction of Standard aerodynamic devices can be found in:

- Masterexcel
  - sheet „Combination Add-ons“ or „Specific Trailer“ column X to BG
- specific_trailer.csv in the Declaration folder,
- Task 2 report



Checkbox to declare applicable standard aerodynamic device(s)
Note that not all combinations are valid → see also:

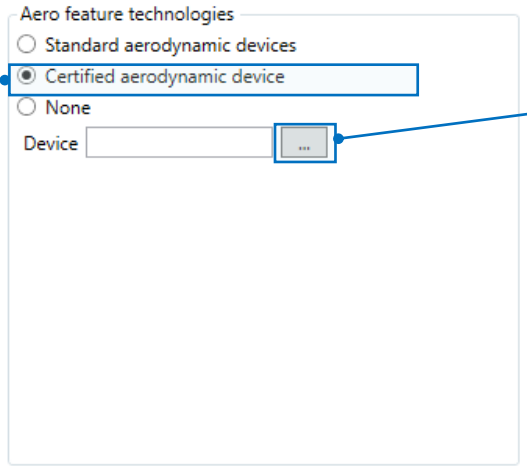
- Masterexcel
  - sheet „Combination Add-ons“
- trailer_combination_addon.csv in the Declaration folder,
- Task 2 report

European Commission

VECTO Trailer Tool – Create or edit trailer job (3/6)

3. Define Aero feature technologies

Choose this if the trailer is equipped with a certified aerodynamic device

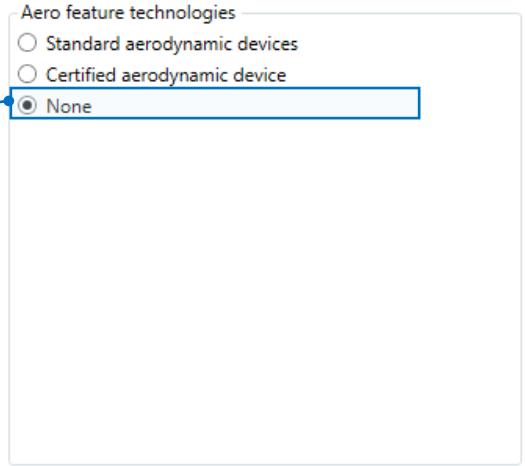


Brows button to search for the XML file of the certified aerodynamic device

European Commission

VECTO Trailer Tool – Create or edit trailer job (4/6)

3. Define Aero feature technologies

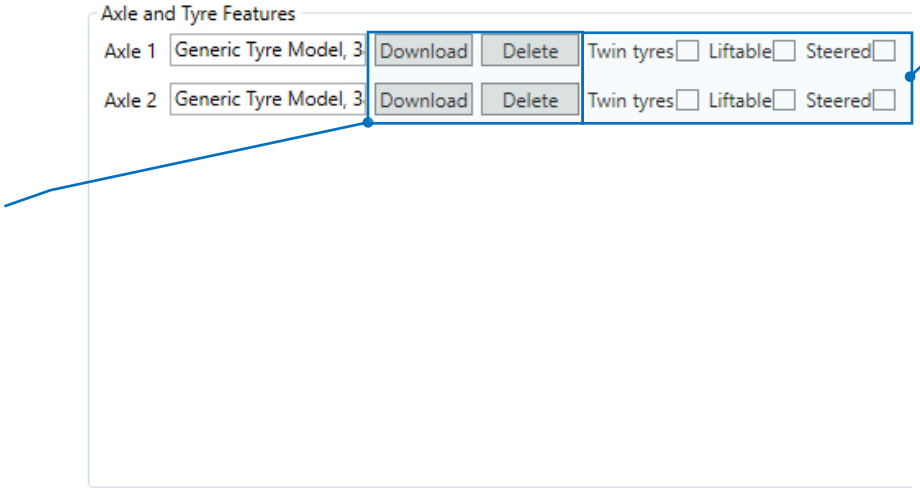


Choose this if the trailer is equipped with no aerodynamic device(s)

European Commission

VECTO Trailer Tool – Create or edit trailer job (5/6)

4. Define Axle and Tyre features



Download
Brows button to search for tyre XMLs

Delete
Delete loaded or pre defined tyre XML

Checkbox to declare applicable tyre/axle features

Effect of liftable/steered axles can be found in:

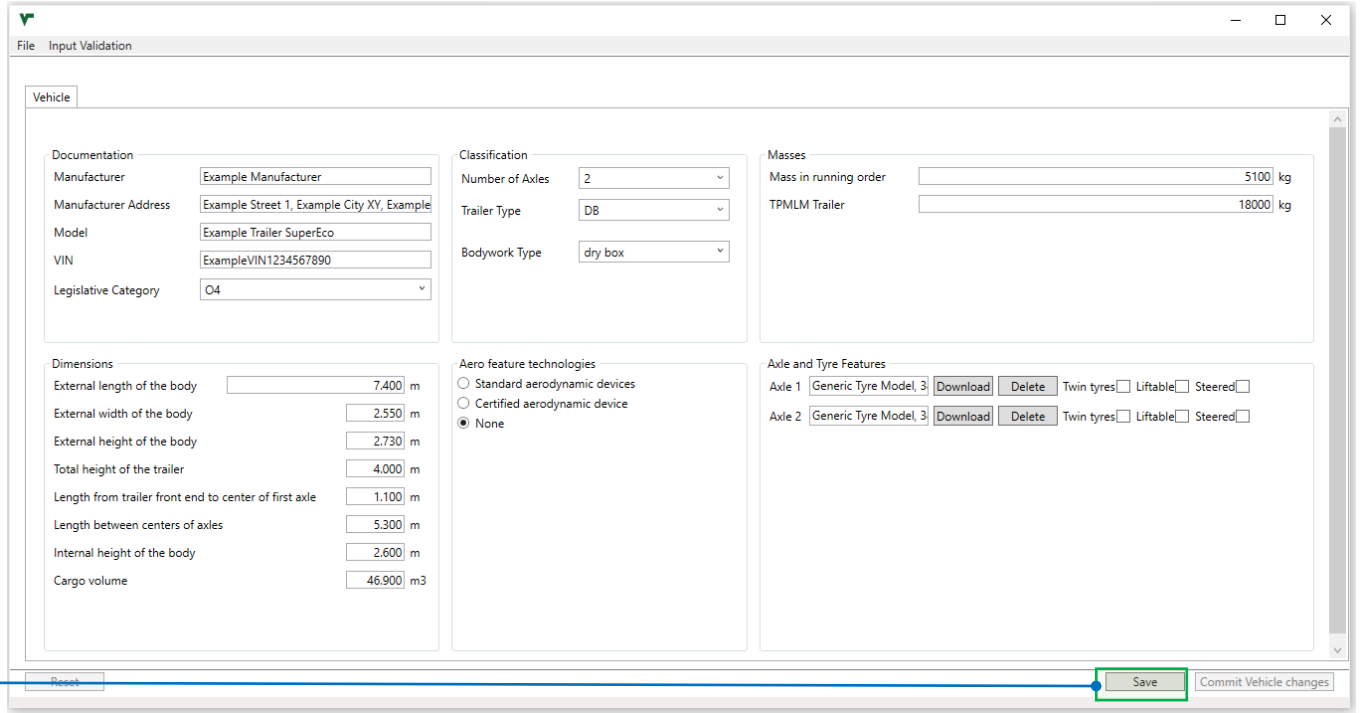
- Masterexcel
  - sheet „Specific Trailer“ column E to P
- specific_trailer.csv in the Declaration folder,
- or the Task 2 report

European Commission

VECTO Trailer Tool – Create or edit trailer job (6/6)

5. Save Job file

6. Press “Start” in Job file editor to start simulation of active Jobs



When the Job-file is saved, a message window will pop up if any errors are detected in the inputs

European Commission

Generation of input XML for a certified aero device (1/X)

- The XML needs to be created by the supplier within a separate certification process
- A template of the XML is located in the downloadable VECTO Trailer Tool package in the “Generic vehicles” folder
- The XSD file is located in the downloadable VECTO Trailer Tool package in the “XML” folder

European Commission

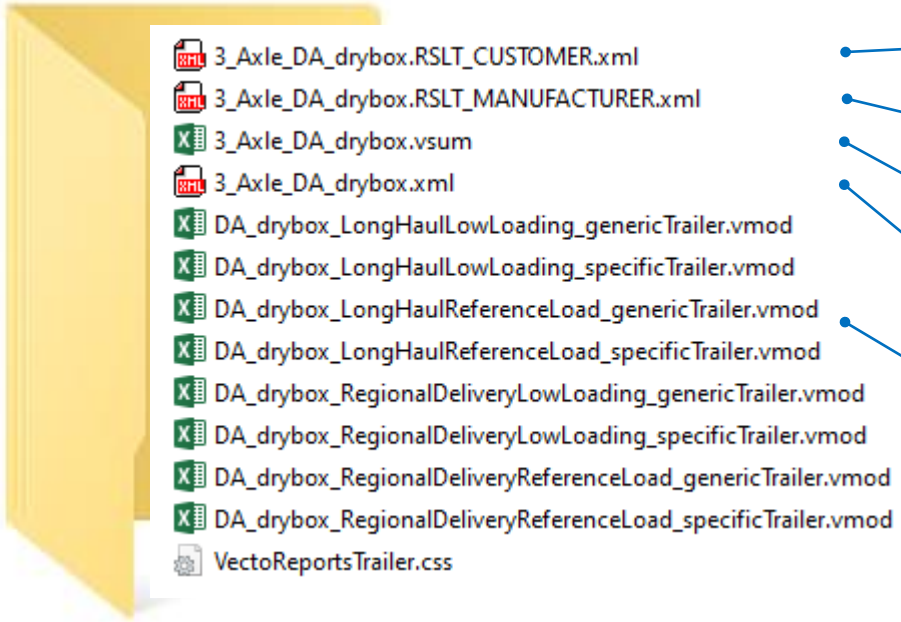
Generation of input XML for a certified aero device (2/X)

- Snapshot XML vor hashing
- Erklären wie hashing funktioniert + Snapshot hashing tool
- Snapshot fertiges XML

European Commission

VECTO Trailer Tool – Result files (1/2)

Outputs created by the VECTO Trailer Tool including the corresponding Job file



- CIF/ Customer Information File
- MRF/ Manufacturers Records File
- vsum File
- Job-File
- vmod File per mission and payload combination for the reference and specific trailer

European Commission

VECTO Trailer Tool – Results (2/2)

Result file	Description
Manufacturer's Records File (MRF)	**Output for regulatory purpose in XML format** ref to definition in the regulation? 'manufacturer's records file' means a file produced by the simulation tool which contains manufacturer related information, a documentation of the input data and input information to the simulation tool and the results for CO2 emissions and fuel consumption;
Customer Information file (CIF)	**Output for regulatory purpose in XML format** ref to definition in the regulation? 'customer information file' means a file produced by the simulation tool which contains a defined set of vehicle related information and the results for CO2 emissions and fuel consumption as defined in Part II of Annex IV;
Pdf-output	Formatted printable output of the customer information file
vsum-file*	Output for engineering purposes in csv-format. Single line of results per each simulated combination of trailer, mission profile and payload with aggregate and or average values
vmod-file*	Output for engineering purposes in csv-format. Single file per each simulated combination of trailer, mission profile and payload with time-resolved simulation results from the VECTO core.

* A documentation of the results contained in the vsum and vmod files can be found in the help file for the VECTO calculation core (help.html), which is also included in the VECTO Trailer Tool release.

European Commission

Overview content of Masterexcel (1/4)

The VECTO Trailer Tool Masterexcel is an integral part of the tool documentation and fulfils the following functions:

- Documentation of the classification matrix used internally in the tool, which is more detailed than the classification according to Annex I.
- Documentation of all generic data stored in the tool (e.g. mission profile and payload allocation, specifications of reference trailers, standard values for reduction rates of aerodynamic devices etc.)
- Documentation of the formulas for calculating the air drag for the various combinations of generic towing vehicles and trailers
- Documentation of bonus factors for liftable and steered axles

The content of the individual sheets is described in overview on the next slides.

In the context of a normal application of the tool, it is not necessary to engage with the Masterexcel.

European Commission

Overview content of Masterexcel (2/4)

Excel sheet	Description
Classification	General information stored in the tool for each trailer configuration • Vehicle group acc. to Annex 1 • Information on the allocated generic towing vehicle • Mission allocation + weighting for aggregated results • Payload and axle weight shares per mission
Generic CAD vehicle	Main dimensions and air drag values of the generic CAD models used as a starting point to calculate the air drag of trailers with different dimensions (could be the reference- or the specific trailer)
Reference Trailer	Information on the reference trailers for each trailer configuration • Curb mass and cargo volume • Tyre specifications • Main external dimensions to calculate the air drag • Indication on which aero corrections are applied to which trailer configuration • „1“ means aero correction is applied • „0“ means aero correction is not applied

European Commission

Overview content of Masterexcel (3/4)

Excel sheet	Description
...	...
Specific trailer	Information on technologies that may be present on a specific trailer • Bonus factors for liftable and steered axles depending on trailer configuration, mission and payload • These factors are to be understood as reduction factors on overall fuel consumption / CO_2 • Indication on which aero corrections are applied to which trailer configuration • „1“ means aero correction is applied • „0" means aero correction is not applied *The individual aero correction formulas can be viewed in the “Reference/Specific Trailer Aero" sheet and are documented in detail in Task 2 report section 2.4.5.* • Aero reduction values for standard aerodynamic devices acc. to sheet „Combination Add-ons DX*“
Standard Trailer	Information on the standard trailers also used in Regulation EU 2017/2400 to calculate the reference factors

*DX... Placeholder depending on trailer type
DA... Semi Trailer
DB... Drawbar Trailer
DC... Centre axle Trailer

European Commission

Overview content of Masterexcel (4/4)

Excel sheet	Description
...	...
Reference (Specific) Trailer Aero	Main dimensions and air drag values of the generic CAD models used as a starting point to calulate the air drag of trailers with different dimensions (Reference/Specific trailer) Step by step calculation process on how to apply the individual aero corrections to get the final air drag values for a reference/specific trailer In the columns AN to AX, the aero-relevant data can be entered for the individual trailer groups. The results for CdxA(0) as well as the polynomial coefficients of the polar can then be found in the columns DT and EB to ED.
Combination Add-ons DX*	Air drag reduction in % for the specific combination of standard aerodynamic devices (the reduction values can also be found in the sheet „Specific Trailer“
Efficiency Ratios - DA	Example on how the individual and weighted efficiency ratios are calculated based on a DA trailer

*DX... Placeholder depending on trailer type
DA... Semi Trailer
DB... Drawbar Trailer
DC... Centre axle Traielr

European Commission